SemCrys: Toward a Tokenizer-Native Semantic Substrate for Machine Cognition

Semantic Crystallization, Concept-Native Representation, and the Reorganization of Cognitive Compression Around Meaning Topology

Reconstructing the Relationship Between Representation, Semantic Stability, and Machine Reasoning Through Stabilized Semantic Symbol Systems

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Abstract

Large language models currently operate over tokenization systems optimized primarily around statistical recurrence of lexical surfaces rather than stabilized semantic structures. As a result, semantically recurring concepts, phrases, and higher-order abstractions remain fragmented across probabilistic lexical decompositions, imposing persistent reconstruction overhead on reasoning systems operating within finite context windows.

This paper introduces *SemCrys* (Semantic Crystallization), a tokenizer-native semantic representation framework in which recurrent semantic structures stabilize into canonical representational units integrated directly into the tokenizer and embedding substrate of the model itself. Unlike session-local symbolic substitution systems such as SemCrys0, SemCrys proposes a cooled semantic layer in which frequently recurring semantic structures become native representational objects requiring no active legend negotiation during ordinary operation.

SemCrys reframes semantic compression as a representational-topological problem rather than solely a token-frequency optimization problem. The framework draws historical lineage from Leibniz's *characteristica universalis*, pasigraphy, lexicography, philosophical language systems, and symbolic representation traditions, while diverging from them through adaptive semantic stabilization and machine-native semantic crystallization.

The paper formalizes semantic cooling dynamics, semantic atoms, semantic recurrence economics, tokenizer-native semantic topology, adaptive versus stabilized symbolic layers, semantic governance requirements, and concept-native reasoning surfaces. It further argues that semantic-native tokenization constitutes

a new class of cognitive substrate engineering capable of increasing semantic density, continuity stability, and representational efficiency without requiring proportional architectural scaling.

SemCrys does not assume that improved representation alone guarantees improved reasoning performance. Rather, it proposes that reducing semantic reconstruction overhead and increasing representational stability may enable more efficient utilization of existing reasoning capacity within finite computational and contextual constraints.

The central claim of SemCrys is that machine cognition may be materially improved when representational substrates are reorganized around stabilized meaning topology rather than purely statistical lexical recurrence.

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1. Introduction

Large language models currently process language through tokenization systems optimized primarily around statistical recurrence of character sequences and lexical surfaces.

This creates a fundamental mismatch between:

- representational organization, and:
- semantic organization.

Conceptually central ideas often remain fragmented across multiple token surfaces despite recurring continuously throughout reasoning processes.

Current approaches to extending effective reasoning capacity typically focus on:

- architectural scaling,
- retrieval augmentation,
- recurrent memory systems,
- sparse attention,
- or context-window expansion.

SemCrys explores a different axis entirely:

- semantic-native representation.

Rather than increasing memory capacity directly or compressing prompts externally, SemCrys investigates whether representational substrates themselves can be reorganized around stabilized semantic recurrence structures.

The core intuition is simple:

> concepts repeatedly reconstructed during reasoning should eventually become native representational objects.

SemCrys therefore proposes semantic crystallization:

- the stabilization of recurring semantic structures into canonical symbolic representations integrated directly into tokenizer and embedding substrates.

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2. Historical Lineage

2.1 Leibniz and the Characteristica Universalis

The deepest historical ancestor of SemCrys is Gottfried Wilhelm Leibniz's proposed *characteristica universalis*:

- a universal symbolic language intended to represent concepts through formal symbolic structure.

Leibniz envisioned systems in which reasoning could operate directly over stabilized symbolic meaning structures rather than ambiguous natural-language surfaces.

SemCrys may be understood as a machine-native descendant of this lineage:

- not a universal human language, but:
- a semantic-native representational substrate for machine cognition.

Unlike Leibniz's proposal, however, SemCrys emerges through adaptive semantic recurrence within machine learning systems rather than entirely through hand-constructed ontological specification.

2.2 Pasigraphy and Universal Symbol Systems

Pasigraphic systems attempted to create symbolic representations understandable across linguistic boundaries.

SemCrys shares structural affinities with pasigraphy:

- concept-oriented notation,
- symbolic semantic representation,
- and abstraction from phonetic surfaces.

However, SemCrys differs fundamentally in purpose.

Pasigraphy primarily sought:

- cross-human linguistic interoperability.

SemCrys instead seeks:

- semantic density amplification,
- tokenizer-native semantic stabilization,
- and representational efficiency for machine cognition.

Its primary consumer is therefore not the human reader, but the reasoning substrate itself.

2.3 Philosophical Languages and Semantic Taxonomy

Philosophical language projects by Wilkins, Dalgarno, and related thinkers attempted to construct rational symbolic taxonomies capable of organizing all knowledge into coherent semantic systems.

These projects frequently encountered rigidity problems:

- semantic evolution,
- conceptual drift,
- taxonomic brittleness,
- and representational ossification.

SemCrys diverges by allowing semantic stabilization to emerge adaptively from semantic recurrence patterns rather than requiring complete a priori ontological specification.

In SemCrys:

- semantic structures cool into canonicalization through recurrence and stability, rather than:
- being exhaustively imposed beforehand.

2.4 Lexicography and Semantic Stabilization

SemCrys also inherits from lexicography.

Traditional dictionaries stabilize:

- lexical surfaces.

SemCrys instead stabilizes:

- semantic recurrence structures.

These may include:

- word senses,
- phrases,
- conceptual operators,
- proper nouns,
- semantic relations,
- and higher-order abstractions.

The SemCrys lexicon therefore functions simultaneously as:

- tokenizer substrate,

- semantic identity layer,
- and representational topology map.

2.5 Trithemius, Artificial Alphabets, and Symbolic Encoding

Johannes Trithemius and related symbolic encoding traditions explored artificial alphabets, symbolic writing systems, and transformational encoding frameworks.

SemCrys shares structural characteristics with symbolic encoding systems:

- artificial symbolic surfaces,
- governed mappings,
- and semantic compaction.

However, SemCrys is not steganography.

Its purpose is not concealment.

Instead, SemCrys seeks:

- semantic stabilization,
- representational efficiency,
- and concept-native cognition.

Where steganographic systems obscure meaning, SemCrys seeks to make semantic structure more computationally operational and representationally efficient.

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3. SemCrys0 vs SemCrys

3.1 SemCrys0

SemCrys0 operates through:

- session-local legend injection,
- adaptive symbolic substitution,
- and temporary semantic negotiation inside active context windows.

It demonstrates that semantic stabilization can emerge from communication-protocol engineering alone, without architectural modification of the underlying model. It is important, however, to state precisely what this does and does not provide. On a conventional (for example, byte-pair) tokenizer, SemCrys0's inline symbolic surface does not reduce token count; typically it increases it. Frequently used words are already encoded as single tokens, whereas novel symbolic surfaces (rare glyphs, canonical identifiers, digit strings) fragment into multiple tokens, and any in-context legend adds further tokens. SemCrys0's contribution is therefore representational rather than compressive: it stabilizes recurring meaning into reusable symbolic identity and serves as the discovery frontier from which durable atoms are identified. Token-level efficiency appears

only once stabilized atoms are integrated into the tokenizer and embedding substrate and the model is fine-tuned to operate over them (Sections 3.2 and 6).

SemCrys0 is:

- adaptive,
- volatile,
- session-local,
- legend-dependent,
- and orchestration-oriented.

3.2 SemCrys

SemCrys extends the SemCrys0 insight into the representational substrate itself.

Frequently recurring semantic structures become:

- tokenizer-native,
- embedding-native,
- stabilized semantic atoms.

Legends remain necessary only for:

- rare concepts,
- experimental abstractions,
- session-specific terminology,
- emergent semantic frontiers,
- or unstable conceptual regions.

SemCrys therefore functions as:

- a semantic cooling system, where:
- recurrent semantic structures crystallize into native representational forms.

3.3 Semantic Thermal States

SemCrys introduces semantic thermal states:

- Hot — adaptive, provisional, volatile
- Warm — reusable but mutable
- Cold — stabilized semantic identity
- Frozen — tokenizer-native canonical representation

This creates a semantic metabolism model in which concepts evolve from temporary symbolic negotiation into stabilized representational infrastructure.

This forms an adaptive–canonical feedback loop:

- observation,

- promotion,
- integration,
- and reuse.

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4. Semantic Crystallization

4.1 Definition

Semantic crystallization is the process by which recurring semantic structures stabilize into canonical symbolic representations.

These structures may include:

- words,
- phrases,
- concepts,
- proper nouns,
- semantic operators,
- relations,
- or compositional abstractions.

4.2 Semantic Atoms

SemCrys introduces semantic atoms:

- stabilized representational units corresponding to semantic recurrence structures rather than purely lexical fragments.

Unlike conventional tokenizer tokens, semantic atoms optimize for:

- semantic recurrence,
- conceptual coherence,
- representational stability,
- and reasoning reuse.

Operationally, a semantic atom may be implemented as a learned embedding associated with a canonical identifier promoted from recurrent co-activation patterns across token sequences. Promotion criteria may include:

- recurrence frequency,
- contextual stability,
- semantic persistence,
- and mutual information across occurrences.

Semantic persistence may be approximated through:

- cross-context consistency of representations,
- such as low variance in embedding neighborhoods across occurrences.

Semantic atom promotion may therefore be framed as optimizing a tradeoff between:

- representational cost, and:
- semantic reconstruction overhead

under stability and generalization constraints.

Informally, the system seeks to maximize:

- semantic reuse per representational unit

while preserving contextual fidelity.

Promotion and demotion may additionally employ hysteresis thresholds to prevent oscillation of borderline semantic structures between adaptive and stabilized states.

The goal is not merely shorter tokenization, but reduction of repeated reconstruction overhead.

4.3 Concept-Native Representation

Current tokenizers encode:

- statistical lexical recurrence.

SemCrys instead encodes:

- semantic recurrence topology.

For example, a recurring phrase such as:

> "context window extension"

may ordinarily require multiple lexical fragments and attention operations across several tokens.

Under SemCrys, the phrase might eventually stabilize into a canonical semantic atom:

`text id="8p7x2k" [CTX_WINDOW_EXTENSION]`

after sufficient recurrence and contextual stability.

Here
`[CTX_WINDOW_EXTENSION]` is notation for a single *native* vocabulary unit (a learned token and embedding introduced into the substrate), not a literal bracketed string transmitted to a conventional tokenizer. Written as ordinary text to a stock tokenizer, such a surface form would itself fragment into several tokens; the efficiency claim holds only for the integrated, fine-tuned representation in which the atom is a single token.

Repeated attention over this concept no longer requires reconstruction from multiple lexical fragments.

Rather than repeatedly reconstructing the phrase from distributed token surfaces, the reasoning system may operate over a stabilized semantic unit directly.

This enables:

- concept-native reasoning surfaces,
- reduced reconstruction overhead,
- increased continuity persistence,
- and lower semantic fragmentation.

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5. Semantic Topology and Tokenization

5.1 Statistical Tokenization vs Semantic Tokenization

Conventional tokenizers optimize:

- byte-pair recurrence,
- lexical frequency,
- and statistical character patterns.

SemCrys instead optimizes:

- semantic recurrence,
- concept stability,
- semantic centrality,
- and compositional semantic reuse.

Where BPE and unigram tokenizers optimize compression over character sequences or likelihood over token segmentations, SemCrys optimizes compression over semantic recurrence structures.

This distinction also explains why SemCrys cannot be realized as a surface-level symbol scheme over an unmodified tokenizer. A byte-pair tokenizer is already a frequency-optimized compressor of lexical surfaces: it encodes common words as single tokens and fragments unfamiliar symbolic surfaces into many. Introducing novel symbol surfaces therefore tends to increase token count rather than reduce it. Semantic-native efficiency is consequently a substrate-integration property rather than an inline-encoding technique, requiring vocabulary and embedding extension and fine-tuning (Section 6). SemCrys therefore treats tokenizer modification as necessary rather than optional.

SemCrys may therefore be viewed as:

- a form of dynamic vocabulary learning driven by semantic recurrence rather than byte-level statistics.

5.2 Semantic Topology

SemCrys introduces semantic topology:

- the structured organization of semantic recurrence relationships within representational space.

Semantic topology may be approximated through:

- embedding-space clustering,
- co-occurrence graphs,
- attention-derived connectivity patterns,
- recurrent semantic activation structures,
- and attention co-activation statistics across heads and layers,

where stable, high-density regions in representational space correspond to candidate semantic atoms and persistent co-attention patterns indicate candidate semantic cohesion.

The representational substrate therefore becomes organized around semantic relational structure rather than solely lexical statistics.

5.3 Semantic Compression

SemCrys does not merely compress representational surfaces.

It compresses:

- semantic recurrence itself.

A stabilized semantic atom may represent:

- an idea,
- a relation,
- a phrase,
- or a conceptual operator

with dramatically reduced reconstruction overhead.

5.4 Semantic Density Amplification

The effective semantic payload per representational unit increases when recurrent semantic structures become native representational objects.

This may improve:

- continuity persistence,
- semantic coherence,
- abstraction stability,
- and context efficiency

without requiring proportional increases in context-window size.

SemCrys does not assume that improved representation alone guarantees improved reasoning capability. Rather, it proposes that reducing reconstruction overhead and increasing semantic stability may allow more efficient utilization of existing reasoning capacity.

The system must also balance:

- semantic granularity, against:
- representational efficiency.

Excessive semantic atom proliferation reintroduces fragmentation, while insufficient stabilization preserves reconstruction overhead.

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6. Prototype Implementation Pathway

A minimal SemCrys prototype pathway may proceed incrementally:

1. Begin with a conventional tokenizer and embedding system.
2. Track recurrent n-gram structures and embedding-space semantic clusters during training or inference.
3. Measure recurrence frequency, contextual stability, semantic persistence, and cross-context representational consistency.
4. Promote stable semantic clusters into candidate semantic atoms once thresholds are exceeded.
5. Assign canonical identifiers to promoted semantic atoms.
6. Inject promoted atoms into tokenizer vocabulary and embedding space.
7. Fine-tune the model to operate over mixed lexical and semantic-native representations.
8. Preserve adaptive symbolic frontier mechanisms for unstable or session-local concepts.

Semantic atom discovery may occur during:

- pretraining,
- fine-tuning,
- or online inference logging,

with later consolidation into tokenizer vocabularies and embedding substrates.

Promoted semantic atoms may additionally be appended to existing vocabularies without invalidating prior tokenization behavior, enabling backward-compatible deployment pathways.

This pathway does not require complete replacement of existing tokenization infrastructure. SemCrys may instead emerge incrementally through layered semantic-native augmentation.

Evaluation may include:

- token count reduction for recurring concepts,
- perplexity or loss changes under mixed vocabularies,
- semantic continuity measures across long contexts,
- and reconstruction-overhead reduction metrics.

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7. Governance and Semantic Constitutionalization

7.1 Semantic Governance Problem

Once semantic atoms become tokenizer-native, representational substrate design becomes a form of cognitive governance.

Whoever controls:

- canonical semantic bindings,
- stabilization criteria,
- tokenizer-native concepts,
- and semantic compression priorities

influences:

- cognitive ergonomics,
- semantic accessibility,
- conceptual gravity,
- and epistemic navigation.

7.2 Semantic Constitutionalization

SemCrys therefore requires constitutional semantic governance.

Semantic stabilization must remain:

- governed,
- reviewable,
- reversible,
- and admissibility-scoped.

Otherwise semantic-native tokenization risks becoming:

- epistemically manipulative,
- semantically centralized,
- or cognitively distortive.

7.3 Manipulative Engagement Loops

Semantic-native representational systems must resist:

- semantic salience inflation,
- engagement-optimized semantic gravity,
- emotional-priority escalation,
- and representational steering.

Substrate-level semantic compression creates substrate-level influence power.

Governance therefore becomes inseparable from tokenizer design.

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8. Semantic Metabolism

8.1 Semantic Cooling

Concepts stabilize through repeated semantic reuse and contextual persistence.

High-recurrence structures gradually transition from:

- provisional, to:
- canonical.

8.2 Semantic Frontier Regions

Not all semantic structures should stabilize.

Some semantic regions must remain adaptive:

- experimental abstractions,
- emergent terminology,
- local project concepts,
- unstable discoveries,
- and session-specific semantic invention.

SemCrys therefore preserves adaptive frontier zones through symbolic negotiation mechanisms inherited from SemCrys0.

8.3 Adaptive–Canonical Feedback Loop

SemCrys0 functions as the evolutionary frontier of SemCrys.

Session-local symbolic reuse reveals:

- recurring semantic clusters,
- durable conceptual structures,
- high-density semantic operators,
- and reusable abstractions.

These may later crystallize into stabilized semantic atoms.

9. Concept-Native Cognition

9.1 Beyond Lexical Reconstruction

Current reasoning systems repeatedly reconstruct concepts from fragmented lexical surfaces.

SemCrys reduces this burden by introducing:

- concept-native representational surfaces.

9.2 Semantic Operators

Higher-order conceptual structures may themselves stabilize into semantic atoms.

Examples include:

- continuity operators,
- semantic governance primitives,
- admissibility structures,
- and abstraction operators.

This creates the possibility of:

- semantic-native reasoning grammars.

9.3 Cognitive Implications

Semantic-native representation may improve:

- long-horizon continuity,
- semantic persistence,
- abstraction stability,
- and representational efficiency.

Because transformer attention operates over token sequences, representing recurring concepts as single semantic atoms allows attention mechanisms to operate directly over concept-level units rather than reconstructing them across multiple lexical fragments.

The implication is significant:

> reasoning efficiency depends not only on memory size, but on semantic topology organization within representational substrates.

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10. Limitations

SemCrys introduces substantial unresolved challenges:

- ontology drift,
- stabilization bias,
- canonicalization politics,
- semantic rigidity,
- cognitive lock-in,
- tokenizer compatibility,
- semantic monopolization,
- and substrate-level epistemic influence.

The system also risks:

- over-crystallization,
- semantic ossification,
- suppression of linguistic diversity,
- and representational over-centralization.

SemCrys therefore requires adaptive frontier mechanisms and constitutional governance structures to remain epistemically healthy.

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11. Discussion

SemCrys suggests that future reasoning systems may increasingly resemble:

- semantic operating substrates, rather than:
- probabilistic lexical predictors.

The framework reframes semantic representation itself as:

- cognitive infrastructure.

This shifts part of the scaling problem away from:

- parameter count,
- context-window size,
- and architectural complexity,

toward:

- semantic organization,
- representational topology,
- semantic stabilization,
- and semantic recurrence economics.

SemCrys therefore belongs simultaneously to:

- machine learning,
- lexicography,

- symbolic representation theory,
- cognitive architecture,
- semantic systems,
- tokenizer engineering,
- and philosophy of language.

A concise characterization of the framework is therefore:

> dynamic, semantics-driven vocabulary learning with governed promotion and stability constraints designed to reduce reconstruction overhead in transformer attention systems.

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12. Conclusion

SemCrys proposes that machine cognition may be materially improved when representational substrates are reorganized around stabilized semantic topology rather than purely statistical lexical recurrence.

Emerging from the protocol-layer symbolic encoding mechanisms demonstrated by SemCrys0, SemCrys extends semantic density amplification into tokenizer-native semantic crystallization.

The framework reconnects modern machine cognition to historical efforts toward symbolic universal languages while diverging from them through adaptive semantic stabilization, semantic cooling dynamics, constitutional governance, and machine-native representational topology.

Its central claim is simple:

> meaning itself may become a first-class representational primitive.

Appendix A – SemCrys0 Relationship

SemCrys0 functions as:

- adaptive semantic orchestration,
- session-local symbolic negotiation,
- and semantic frontier exploration.

SemCrys functions as:

- stabilized semantic substrate,
- tokenizer-native semantic cooling,
- and canonical semantic crystallization.

The two systems are complementary rather than competitive.

Appendix B – Proposed Future Directions

- tokenizer-native semantic atom training,
- semantic recurrence metrics,
- semantic stabilization thresholds,
- semantic cooling dynamics,
- adaptive-to-canonical promotion systems,
- constitutional semantic lexicons,
- semantic continuity benchmarking,
- and concept-native reasoning evaluation.

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Related Works

- characteristica universalis

- pasigraphy
- philosophical languages
- semantic compression
- symbolic cognition
- tokenizer-native semantics
- semantic topology
- semantic governance
- semantic tokenization
- concept-native representation
- semantic operating substrates
- symbolic representation systems
- dynamic vocabulary learning
- in-context symbolic encoding
- semantic continuity systems
- semantic crystallization

Corpus-Wide Reference Positioning

This paper argues that large language models currently operate over tokenization systems optimized primarily around statistical recurrence of lexical surfaces rather than stabilized semantic structures. Its nearest external comparison fields are semantic, token, formal. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Statistical compression can improve efficiency while destroying distinctions needed for interpretation; semantic labels can conversely overstate stability that usage does not support. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

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[**E2**] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference*. W3C Recommendation, 18 August 2009. https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] Rico Sennrich, Barry Haddow, and Alexandra Birch. "Neural Machine Translation of Rare Words with Subword Units." *ACL 2016*, pp. 1715-1725. DOI: 10.18653/v1/P16-1162. https://doi.org/10.18653/v1/P16-1162 **Relationship:** Predecessor subword tokenization method used as a contrast to semantically governed token boundaries.

[**E7**] Taku Kudo and John Richardson. "SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing." *EMNLP 2018: System Demonstrations*. DOI: 10.18653/v1/D18-2012.

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Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Governed Semantic Tokens: Tokenization as an Admissibility Boundary.* Ableman Corpus current edition; work identity `PAPER.PUB.18469507`;

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[**A7**] Adam Ableman. *Token Parsimony.* Ableman Corpus current edition; work identity `PAPER.PUB.20382825`;
source SHA-256
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
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DOI: 10.5281/zenodo.20374559